

KRIDHAY MAHESH

[linkedin.com/in/kridhay/](https://www.linkedin.com/in/kridhay/) | github.com/Krldhay/

EDUCATION

Imperial College London

London, UK

MEng Electronic and Information Engineering with a Year Abroad

October 2021 – June 2025

- Grade: First Class Honours
- Selected as the only student in cohort of ~140 to spend my fourth year at the National University of Singapore
- Taught weekly tutorial sessions for the C++ Programming Module, working as an Undergraduate Teaching Assistant

EXPERIENCE

Behaviour Lab (Fintech Startup)

London, UK

Forward-Deployed Software Engineer

September 2025 – present

- Migrated a duplicated, client-specific service into a modular and reusable feature-based architecture using Python, enabling the business to scale and reducing daily developer support time by 8x, saving significant engineering cost
- Collaborated with equity strategists and clients across funds with \$265B+ in AUM to build custom financial signal and data processing algorithms in Pandas, tailored to each fund's weighting rules, benchmarks and team structure
- Delivered a data source feature end-to-end, implementing backend services, database migrations and frontend components using TypeScript and Next.js, to meet client requests for full transparency over their incoming data
- Automated data publishing by designing an asynchronous polling system, deployed on AWS Lambda and integrated into the existing Step Functions pipeline, eliminating a recurring full day manual process
- Coded LLM based agents with authenticated access to internal APIs, dynamically pulling context from our services to automate document filling workflows within the product
- Integrated automated evals using Promptfoo into the CI/CD pipeline, enabling rapid development of our AI features

Forward-Deployed Software Engineer Intern (Industrial Placement)

April 2024 – August 2024

- Designed an alert system with an event-sourced AWS DynamoDB model, ensuring daily execution of critical functions
- Decoupled internal Pydantic/Pandera validation schemas from API responses, isolating core logic from dependencies
- Provisioned an SFTP server using Terraform and AWS Transfer Family, configuring IAM roles and IP traffic rules to enable secure data exchange with client systems

Citi

London, UK

ICG Summer Technology Analyst

June 2023 – August 2023

- Implemented new features for various applications within the Commodities Team using TypeScript, including developing automated tests and calculating performance metrics
- Created a dashboard using Grafana and InfluxDB to analyse the Pricer Application usage statistics, allowing for performance optimisation based on user behaviour

PROJECTS

Interpretable Medical Models Using ML and Symbolic Regression (Thesis)

September 2024 - May 2025

- Implemented novel Symbolic Regression algorithms to develop models for prognosticating outcomes in SICH patients
- Benchmarked developed models against traditional ML techniques and existing clinical scoring systems to validate their competitiveness in terms of interpretability and performance

Optiver Algorithmic Trading Challenge (IC Hack '24)

February 2024

- Implemented Market Making, Arbitrage and Pairs Trading algorithms using Python and Optibook
- Consistently achieved top 3 profit on live exchanges against 30+ teams, and selected for the final to present our strategy

RISC-V Processor and Compiler

December 2022 - March 2023

- Collaborated with a team to develop a pipelined RISC-V 32-bit processor with a reduced ISA using System Verilog
- Built a C90 compiler in C++, implementing parsing, AST construction and code generation to produce RISC-V assembly

ADDITIONAL INFORMATION

Programming Skills: Python, C++, TypeScript, Rust, SQL, SystemVerilog

Technologies: AWS, Next.js, Pandas, Unity, Git, VTune, Terraform, Promptfoo

Leadership: Algorithmic Trading Society (Head of Corporate Relations) AI Furnace (Event Coordinator)

Languages: English (Native), Tamil (Intermediate)

Interests: Weightlifting, Muay Thai, Golf, Guitar